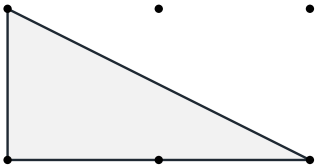
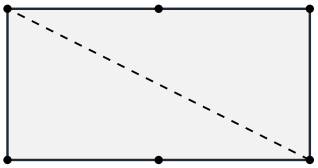
Join dots with straight sides to make one closed outline. Sides cannot cross or touch except at neighboring corners. The area of one grid square is 1. Find or check areas with pieces, cuts or copies.

Input



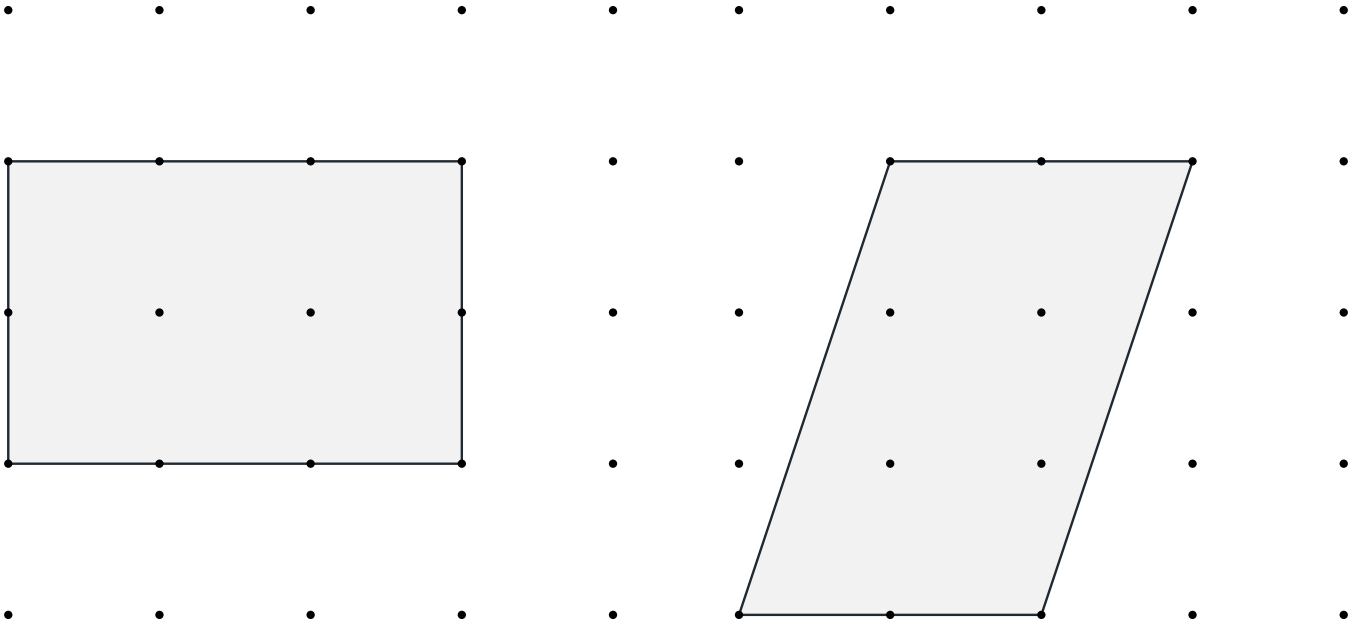
Two copies joined



Output

Rectangle area: 2
One copy's area: 1

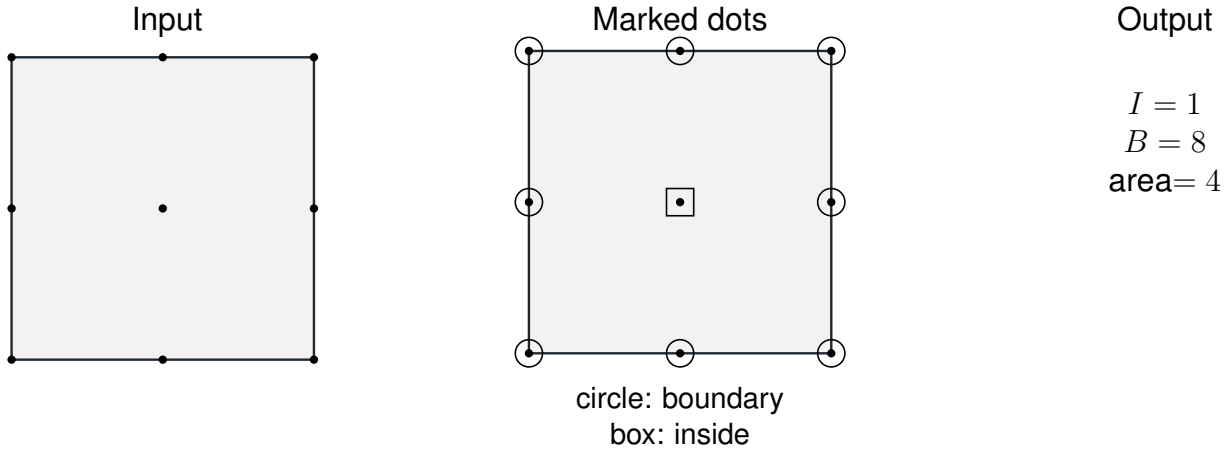
Problem 1: Find the exact area of each shape.



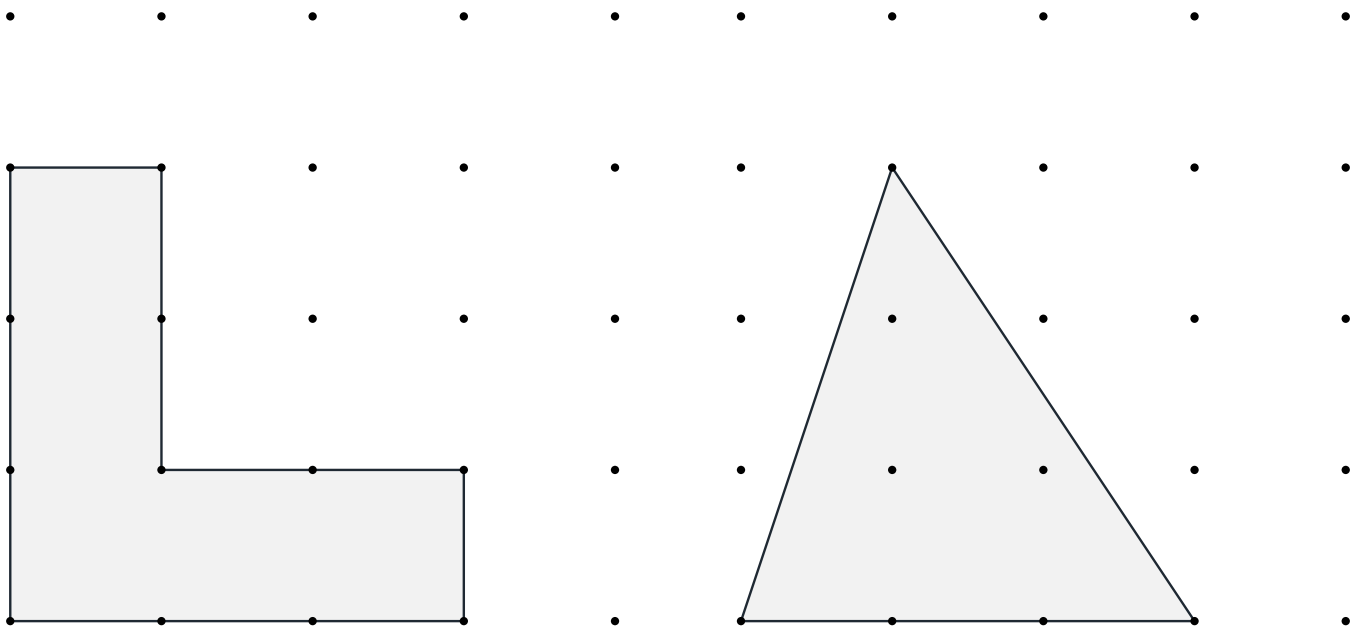
area=_____

area=_____

I counts dots strictly inside. B counts every dot on the outline, including dots between corners. Count each dot once.



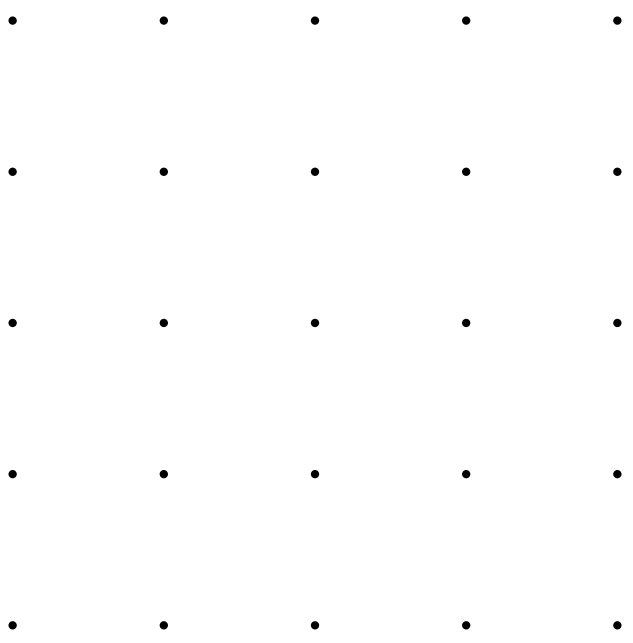
Problem 2: For each shape, mark the dots you count and make an $I, B,$ area record.



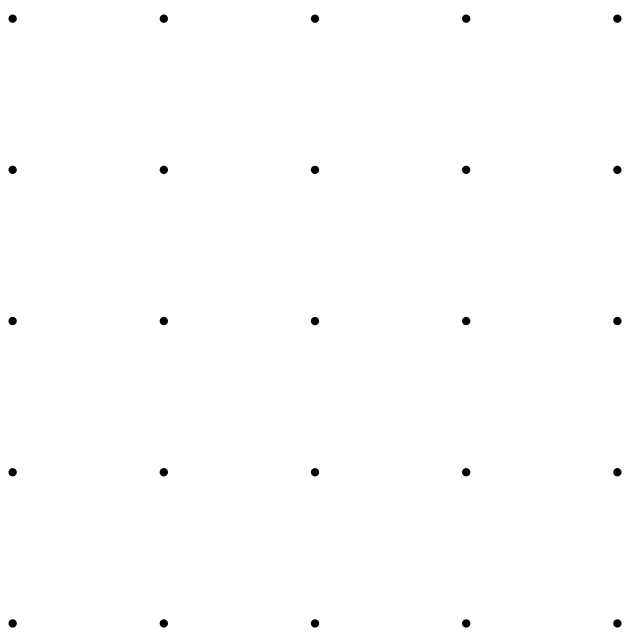
$I =$_____ $B =$_____ area=_____

$I =$_____ $B =$_____ area=_____

Problem 3: Make a triangle and a four-sided shape, each with $I = 2$ and $B = 6$. Find their areas.



$I =$_____ $B =$_____ area=_____



$I =$_____ $B =$_____ area=_____

Problem 4: Find a rule that predicts area from I and B. Use all your records. Draw a new shape that could defeat your rule.



$I =$_____ $B =$_____ area=_____

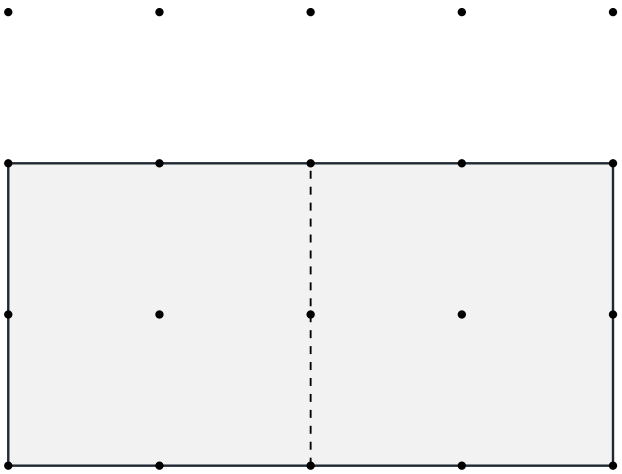
Problem 5: Pick's rule for the allowed polygons is

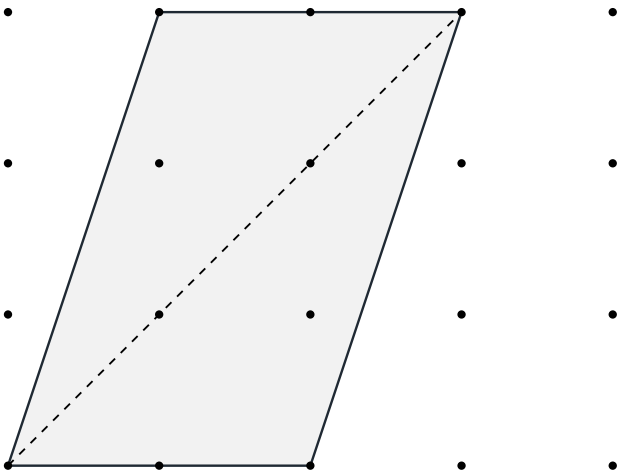
$$\text{area} = I + (\text{half of } B) - 1.$$

Make shapes of area 6 with as many different (I, B) records as you can.



Problem 6: Each dashed seam cuts a shape into two pieces. Make an $I, B,$ area record for each piece and for the whole shape. What changes when the seam disappears?





Left piece:
$I =$_____ $B =$_____ area=_____
Right piece:
$I =$_____ $B =$_____ area=_____
Whole:
$I =$_____ $B =$_____ area=_____

Lower piece:
$I =$_____ $B =$_____ area=_____
Upper piece:
$I =$_____ $B =$_____ area=_____
Whole:
$I =$_____ $B =$_____ area=_____